Krishiv Potluri

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Research Interests

Core Research Interests: My research lies at the intersection of **machine learning**, **computational biology**, and **physics-based simulation**, with a focus on the **computational systems** that make this work **scalable and reproducible**.

Current Directions: **Physics-informed reinforcement learning** for *de novo* antibody design; **generative models** for protein and biomaterial design; **energy-based models** of cancer cell states; **leakage-audited benchmarks** for genomic prediction; and **efficient, reproducible systems** for scientific computing.

Research Experience

Oden Institute for Computational Engineering & Sciences, UT Austin · *12 hrs/wk, 40 wks/yr* — *Austin, TX*

Student Researcher, Computational Visualization Center, advised by Prof. Chandrajit Bajaj — *Mar. 2025 – Present*

- **Antibody Design**: Developing a port-Hamiltonian reinforcement learning model that generates *de novo* antibodies by optimizing transition paths across an energy landscape, rather than screening existing libraries.
- **RNA Therapeutics**: Recoding circular RNA to translate reliably as a durable therapy for the lab's Parkinson's disease research; presented at the 2026 Demo Day.

De Novo Protein-Inhibitor Design for Antibiotic Resistance · *21 hrs/wk, 32 wks/yr* — *Independent Research*

First Author · Under Review, RECOMB 2027 · Regeneron ISEF Finalist. [Preprint] / [Code] — *Jun. 2024 – Jan. 2025*

- **Problem**: TEM-171 β-lactamase lets bacteria destroy most common antibiotics, and designing a protein inhibitor normally takes years of wet-lab iteration.
- **Approach**: Built an AI pipeline that generates inhibitor proteins and validates binding with deep-learning structure prediction, steered molecular dynamics, and free-energy calculations on A100/H100 GPUs.
- **Result**: **94%** of designs formed stable interfaces (ipTM > 0.8) at **38.4 s** each; the pipeline is target-agnostic, turning a multi-year design cycle into a compute problem.

AI Inverse Design of Cardiac-Repair Biomaterials · *20 hrs/wk, 40 wks/yr* — *Independent Research*

First Author · Accepted, BIBM 2026 · 1st Place, TXSEF 2026. [Model Code] / [Simulation Code] — *May 2025 – Feb. 2026*

- **Problem**: After a heart attack, scar permanently replaces muscle, and matching a regenerative hydrogel to one patient's heart takes years of trial and error.
- **Approach**: Pretrained **HYDRA-BERT** (115M parameters, 3.1M+ samples, 16 A100s), the first chemical language model applied to cardiac therapy, and built 10 algorithms that turn clinical MRI into patient-specific heart simulations with 3.66M elements.
- **Result**: Clinically meaningful improvement in cardiac function across all patients tested, compressing years of design iteration into days.

Energy-Landscape Modeling of Acute Myeloid Leukemia — *Independent Research*

First Author, with M. Yurukcu · Accepted, ICBBS 2026. [Code] — *2026*

- **Problem**: Single-cell methods bury the rare leukemic stem cells that drive relapse, and existing atlases name cell states without measuring how stable they are.
- **Approach**: Modeled cell density as a Boltzmann distribution and trained a score-based network to recover the energy landscape, giving **79 cell states** physical fingerprints (basin depth, first-passage time, committor).
- **Result**: Recovered known drug responses with no supervision (MEK/HDAC, $\rho = -0.42$; BCL2), with all 40 evaluable signatures replicating in an independent **748,679-cell** atlas.

Benchmarking Personal-Genome Expression Prediction — *Independent Research*

First Author, with M. Yurukcu · Under Review, ICCBB 2026. [Code] — *2026*

- **Problem**: DNA sequence models predict which genes are expressed but fail at which people express them more, the axis that links a genome to disease risk.
- **Approach**: Built a leakage-audited benchmark on **449 individuals** with per-gene heritability ceilings, gene splits by whole chromosome, and a 251,275-pair paralog audit.
- **Result**: Per-gene models reach a median correlation of **0.23** on unseen individuals versus **0.05** for a pretrained sequence model; localized the gap to the learned map, not the fitting.

Southern Methodist University *Dallas, TX*

Research Placement with Farhad Azadjoutabari, awarded for 1st Place at TXSEF 2026 *2026*

- Worked on a team building a county-wide machine-learning model of crash occurrence in Dallas County, shifting traffic-safety planning from reactive to predictive.

Publications (all first-author)

Conference Proceedings

[C2] AI-Driven Inverse Design of Bio-Instructive Hydrogels for Myocardial Regeneration.
Krishiv Potluri
IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 2026 (Accepted; 1st Place, TXSEF 2026) [Link]

[C1] An Empirical Energy Landscape of Acute Myeloid Leukemia: Cell-State Biophysics and Drug Action as Landscape Perturbation.
Krishiv Potluri, M. Yurukcu
International Conference on Biophysics and Biomedical Sciences (ICBBS), 2026 (Accepted)

Under Review

[U2] De Novo Structure-Based Design of TEM-171 β-Lactamase Protein Inhibitors Using Integrated Deep Learning and Multi-Scale Simulations to Combat Bacterial Resistance.
Krishiv Potluri
International Conference on Research in Computational Molecular Biology (RECOMB), 2027
bioRxiv, 2025 (Regeneron ISEF 2025 Finalist) [Paper]

[U1] A Leakage-Audited Benchmark for Cross-Individual Gene Expression Prediction and a Dissection of the Held-Out-Gene Transfer Gap.
Krishiv Potluri, M. Yurukcu
10th International Conference on Computational Biology and Bioinformatics (ICCBB), 2026

Honors and Awards

2025 **Regeneron ISEF Finalist**, one of nearly 1,700 finalists from 62 countries, regions, and territories [Link]

2026 **1st Place, Computational Biology & Bioinformatics**, Texas Science & Engineering Fair; scholarship and invitation to the Governor's Science & Technology Champions Academy [Link]

2025 **Davidson Fellows Scholarship Finalist**, one of 33 finalists nationwide and the only one from Texas [Link]

2025 **1st Place Biochemistry & Best of Fair**, Greater Austin Regional Science & Engineering Fair, with the Texas Tech University Best Abstract Award [Link]

2026 **2nd Place, Computational Biology**, Greater Austin Regional Science & Engineering Fair [Link]

2024 **Recursion RxRx Prize**, Lux Capital Bio $\times$ ML Hackathon, for the best use of Recursion's RxRx datasets [Link]

2024–25 **USA Biology Olympiad Semifinalist**, Certificate of Merit, twice; top 225 of 12,000 (top 1.9%)

2025 **NSDA Degree of Special Distinction**, top 7.5% of National Speech & Debate Association members

2025 **TFA State Qualifier, Public Forum**, one of 98 teams statewide, with multiple top-speaker awards

2023–26 **VEX World Championship Qualifier**, competitor (2023, 2024) and coach (2026); peak world rank 14th

2026 **National Merit Commended Student**, top 4% of over 1.2 million PSAT/NMSQT entrants

Education

BASIS Cedar Park — *Cedar Park, TX*

High School Diploma, Expected May 2027 — *Aug. 2023 – May 2027*

GPA 4.86 / 5.00 weighted · 4.00 / 4.00 unweighted

ACT **35** Composite · **35** Superscore (Math **36**, Reading **35**, English **35**, Science **34**)

AP Exams

Calculus AB **5**	Computer Science A **5**	English Language **5**	U.S. History **5**
Calculus BC **5**	CS Principles **4**	English Literature **5**	U.S. Government **5**
Biology **5**	Chemistry **4**	Psychology **5**	World History **5**

Coursework Linear Algebra, Multivariable Calculus, Honors Physics, AP Spanish Language; honors capstones in Biotechnology, Spanish Literature & Culture, and Literary Genres; Senior Project (High Honors)

Professional Experience

Shire Bio · *15 hrs/wk, 11 wks/yr* — *Austin, TX*

ML & Bioinformatics Intern, Production Engineering — *Jun. 2025 – Aug. 2025*

- Engineered and containerized **80+ genomics tools** for the startup's genomic-analysis platform, creating the reasoning layer that lets its AI models compose and execute deep, multi-step workflows.
- Shipped machine-learning tools into the production bioinformatics platform that company scientists use daily.
- Developed AI-driven research workflows, effectively an agentic development environment for bioinformatics pipelines, on AWS, Docker, and Git.

Austin Energy · *20 hrs/wk, 12 wks/yr* — *Austin, TX*

Data Science & Analytics Intern, Data Office — *Jun. 2024 – Aug. 2024*

- Analyzed consumption data across **500,000+ accounts** at the nation's 8th-largest publicly owned electric utility, contributing to measurable gains in operational efficiency.
- Built Python demand forecasts and data algorithms that turned complex energy metrics into decisions reviewed by senior management and used by city leadership.

Austin Police Department — *Austin, TX*

Intern — *Aug. 2025*

- Designed community-engagement and youth-recruitment models to strengthen relationships between the department and local youth.
- Analyzed community-policing initiatives with department leadership and presented recommendations grounded in research on police–community relations.

Leadership & Mentorship

VEX Robotics, BASIS Bots · *10 hrs/wk, 38 wks/yr* — *Cedar Park, TX*

Head Coach (Team 2668B) & Competitor-Programmer (Team 2668A) — *2022 – Present*

- **Coach**: Led a middle-school VEX IQ team to three regional titles, a Texas state championship, two state runner-up finishes, and three World Championship qualifications.
- **Competitor**: World Championship qualifier in 2023 and 2024, with a peak world ranking of **14th** (62nd in 2023) and a world skills ranking of 246.

Speech & Debate, NSDA and Texas Forensic Association · *12 hrs/wk, 36 wks/yr* — *Cedar Park, TX*

Varsity Public Forum Captain & Novice Coach — *2023 – Present*

- **First in school history** to qualify for the TFA State Championship, one of 98 teams statewide, earning multiple top-speaker awards and the NSDA Degree of Special Distinction.
- Coached **20+ novices** through the full case-writing process (contention structure, impact scenarios, and internal link chains) into skilled varsity debaters.

Biotechnology Club, BASIS Cedar Park · *6 hrs/wk, 35 wks/yr* — *Cedar Park, TX*
Founder, President & Mentor — *Sep. 2024 – Present*

- Founded the club to widen access to biotechnology lab work; lead weekly hands-on labs in DNA extraction, tissue culture, and genetic engineering.
- Mentored **50+ students** in core lab techniques and through their first research projects.

Community Service

Key Club International (Kiwanis), BASIS Cedar Park Chapter · *4 hrs/wk, 35 wks/yr* — *Cedar Park, TX*
Vice President & Tutoring Program Founder — *Sep. 2023 – Present*

- Built a student-led tutoring program serving **100+ middle schoolers** in grades 6–8, and tutor middle and high school students across subjects.
- Ran a school-wide book drive of **200+ books** for a low-income Austin-area district, and led food and clothing drives.
- Coordinated partnerships with local nonprofits on service projects and environmental initiatives.

VEX Robotics Summer Camp — *Cedar Park, TX*
Founder & Director — *Summers 2024 – 2026*

- Founded and directed an **eight-week** summer camp, run three summers, teaching robot building, programming, and competition strategy.
- Reached **100+ middle-school students**, opening hands-on robotics to students beyond the competition team.

Athletics

Texans Varsity Baseball & Action Select Travel Baseball · *11 hrs/wk, 44 wks/yr* — *Austin, TX*
Varsity Pitcher & Infielder (RHP / 3B / SS / OF) — *2023 – Present*

- **First freshman in program history** to earn a varsity roster spot.
- Playing select and travel baseball since age nine on elite national teams, competing in tournaments across the country.

Skills

Programming	Python, R, Java, Bash, Git, Linux
Frameworks	PyTorch, CUDA, scanpy, scvi-tools, AnnData, pysam, bcftools, lifelines, scikit-survival
Infrastructure	AWS, Docker, distributed training on 16× NVIDIA A100, A100 / H100 cloud (Brev.dev, Google Colab)
Methods	Chemical language models, reinforcement learning, score-based models, steered MD, eQTL mapping

Hackathons

Morphologic AI, Bio × ML Hackathon (Lux Capital & EvolutionaryScale) — *Virtual*
ML & Cell Segmentation Lead. [Devpost] / [Code] — *Oct. 2024*

- Led machine learning, cell segmentation, and algorithm integration on a team competing against university and industry teams as a high school student.
- Built cell segmentation and morphological readouts that make Recursion's RxRx drug-perturbation images and deep-learning embeddings biologically interpretable.
- Won the **Recursion RxRx Prize** for the best use of Recursion's RxRx datasets, earning a presentation to Recursion's team and a follow-on project with Recursion Pharmaceuticals.

Research Presentations

[T3] Summer Research Demo Day: Circular RNA Design.
Computational Visualization Center, Oden Institute, UT Austin, 2026 (Poster & Demo); shown in the TACC Visualization Lab

[T2] AI-Driven Inverse Design of Bio-Instructive Hydrogels for Myocardial Regeneration.
Texas Science & Engineering Fair, 2026 (1st Place, Computational Biology & Bioinformatics) [Link]

[T1] TEM171 Inhibitor Design via Deep Learning Pipeline.
Regeneron International Science & Engineering Fair, Columbus, OH, 2025 [Link]

Media Coverage

[M6] 2026 Category Awards: Computational Biology & Bioinformatics
Texas Science & Engineering Fair, Texas A&M University, 2026 [Link]

[M5] 2025 Davidson Fellows Scholarship: Finalist
Davidson Institute, 2025 [Link]

[M4] "Central Texas students excel at world's largest science fair, earn top awards, scholarships"
by Allison Miller, CBS Austin, 2025 [Link]

[M3] "Central Texas students won big at the world's top science fair"
City of Austin Office of Sustainability, Facebook, 2025 [Link]

[M2] TEM171 Inhibitor Design via Deep Learning Pipeline
Regeneron ISEF 2025 Project Showcase, 2025 [Link]

[M1] Morphologic AI
Bio × ML Hackathon Project Showcase, Devpost, 2024 [Link]